Nebyat H. Hailu

AI/ML Engineer

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Github

Linkedin

Education

AkiraChix
Diploma In Information Technology
Feb 2025 - Present

Skills

Programming Languages - Python, SQL

Libraries & Frameworks - Pandas, NumPy, NLTK , SpaCy, Scikit-learn, TensorFlow, Torch ,PyTorch, OpenCV, langchain, Hugging Face Transformers, TensorFlow, Keras

Data Visualization - Matplotlib, Seaborn, tqdm

Database Technologies - PostgreSQL, pg vector, Chromadb

Tools & Platforms - Jupyter Notebook, Google Colab, Google AI Developer Kit (ADK), Google Drive FastAPI, Google Cloud and GitHub, Google Gen AI SDK

References

- **Linda Kamau**
  Founder & Executive Director
  AkiraChix
  lkamau@akirachix.com

- **James Mwai**
  Co-Founder & CTO
  Paycloud
  smartemwa@gmail.com

- **Mary Bahati**
  Frontend Engineer
  Temelio
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Personal Statement

How do people learn to trust digital systems around them? This question drew Nebyat to explore how technology can be built responsibly and used for the common good. She believes technology is most powerful when guided by values like accountability and equity. For her, being part of the tech world means engaging fully with both its potential and its responsibility, and she's eager to help shape a future where progress truly serves everyone.

Experience

Recos

AI-powered interview assistant designed to help recruiters identify the right candidates through data-driven insights and skill-mismatch reduction.

- Integrated with Odoo ERP through secure APIs to automatically sync job postings and candidate profiles, enhancing recruiter efficiency while ensuring data privacy.
- Evaluated multiple speech-to-text models and migrated from AssemblyAI to Google Speech-to-Text, achieving significant lower latency, better accent accuracy, and more stable real-time transcription, resulting in higher quality inputs for downstream AI analysis.
- Designed and optimized prompting strategies and integrated the Google Generative AI SDK to generate concise skill summaries and job insights, improving the accuracy, relevance and contextual understanding of candidate-job matches.

Movie Recommendation Agent

A film recommendation system that delivers personalized movie suggestions through semantic search and natural-language understanding.

- Loaded, filtered, and cleaned 1000-movie IMDB dataset with Pandas/NumPy, dropping sparse data for ML readiness, enhanced text quality via lemmatization on titles, genres, and descriptions.
- Built a recommendation pipeline using NLTK/Tiktoken that captured semantic relationships between users and movies using embeddings, enabling context-aware and near-instant retrieval.
- Integrated Google ADK Agent to automate data loading, top-K retrieval, and generate human-like explanations for recommendations.
- Enhanced system reliability with a custom fallback module for natural-language responses, environment-aware caching, and robust exception handling to ensure smooth performance under.

Breast Cancer Image Classification

Developed a deep learning model to classify ultrasound images of breast tissue as benign, malignant, or normal, aiding early detection and diagnosis.

- Automated dataset preparation by ingesting and organizing 1,500+ ultrasound images from Google Drive using Pandas and preprocessing pipelines for efficient training.
- Enhanced model generalization through extensive data augmentation (rotation, zoom, flips, and shears) and transfer learning with MobileNetV2, achieving ~93% validation accuracy—an 11% improvement over baseline CNN models.
- Engineered an end-to-end testing pipeline for preprocessing and classifying unseen scans, mapping probability scores to interpretable clinical labels for practical deployment.